

|   |         |                                                                                                                 |    |     |
|---|---------|-----------------------------------------------------------------------------------------------------------------|----|-----|
| 5 | (a) (i) | Young modulus = stress/strain .....                                                                             | C1 |     |
|   |         | data chosen using point in linear region of graph .....                                                         | M1 |     |
|   |         | Young modulus = $(2.1 \times 10^8)/(1.9 \times 10^{-3})$                                                        |    |     |
|   |         | = $1.1 \times 10^{11}$ Pa .....                                                                                 | A1 | [3] |
|   | (ii)    | This mark was removed from the assessment, owing to a power-of-ten inconsistency in the printed question paper. |    |     |
|   | (b)     | area between lines represents energy/area under curve represents energy ..                                      | M1 |     |
|   |         | when rubber is stretched and then released/two areas are different .....                                        | A1 |     |
|   |         | this energy seen as thermal energy/heating/difference represents energy                                         |    |     |
|   |         | released as heat .....                                                                                          | A1 | [3] |
| 6 | (a)     | <i>either <math>P \propto V^2</math> or <math>P = V^2/R</math></i>                                              | C1 |     |